FLOOD – A NATURAL DISASTER

Floods are the more frequently type natural disasters and occurs when an overflow of water submerges land that is usually dry. Floods are often caused by heavy rainfall, rapid snowmelt or storm surge from a tropical cyclone or tsunami in coastal areas.

CAUSE OF FLOOD

- During times of rain, some of the water is retained in ponds or soil, some is absorbed by grass and vegetation, some evaporates, and the rest travels as surface runoff
- heavy rainfall (locally concentrated or throughout a catchment area), highly accelerated snowmelt, unusual high tides, tsunamis, or failure of dams, severe winds over water, levees, retention ponds, or other structures that retained the water.
- Flooding can be exacerbated by increased amount of impervious surface or by other Floods are caused by many factors or a combination of any of the generally prolonged natural hazards such as wildfires, which reduces the supply of vegetation that can absorb rainfall
- Occur when ponds, lakes, riverbeds, soil, and vegetation cannot absorb all the water
- River flooding also caused by heavy rain, sometimes increased by melting snow.
- Flash flood usually result from intense rainfall over a relatively small area, or if the area was already saturated from previous precipitation.
- Periodic floods occur on many rivers, forming a surrounding region known as the flood plain.

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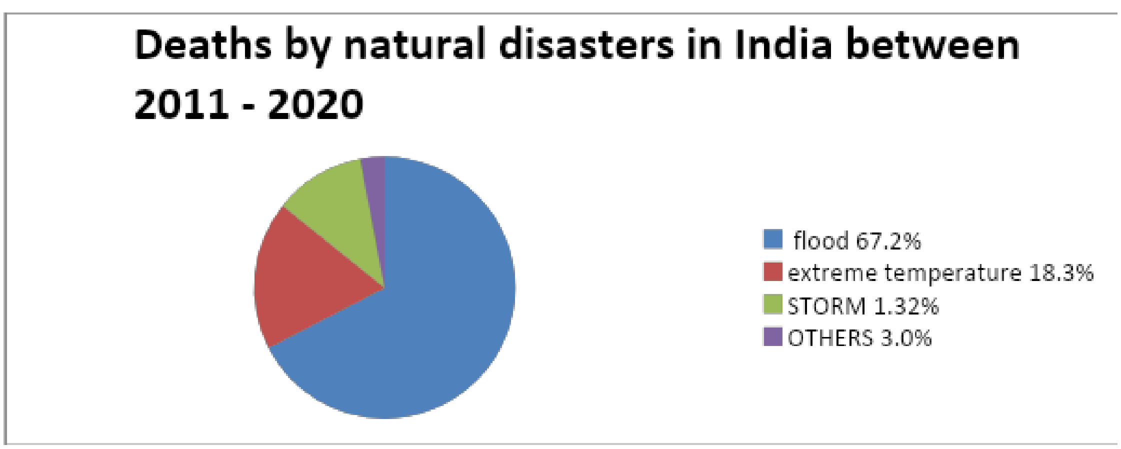
Flooded streets and submerged houses after heavy rainfall, in Warangal district on Sunday.(PTI)

EFFECT OF FLOOD

- Damages property and endangers the lives of humans and other species
- Soil erosion
- Concomitant sediment deposition
- Structural damages such as bridge abutments, bank lines, sewer lines, and other structures within floodways.
- Spawning grounds for fish and other wildlife habitats can become polluted or completely destroyed.
- Some prolonged high floods can delay traffic in areas which lack elevated roadways.
- Flood can interfere with drainage and economic use of lands, such as interfering with farming.
- Waterway navigation and hydroelectric power are often impaired.
- Financial losses due to floods are typically millions of dollars each year, with the worst floods in recent U.S. history having cost billions of dollars.



METHODS OF FLOOD MANEGEMENT

Some methods of flood control have been practiced since ancient times.

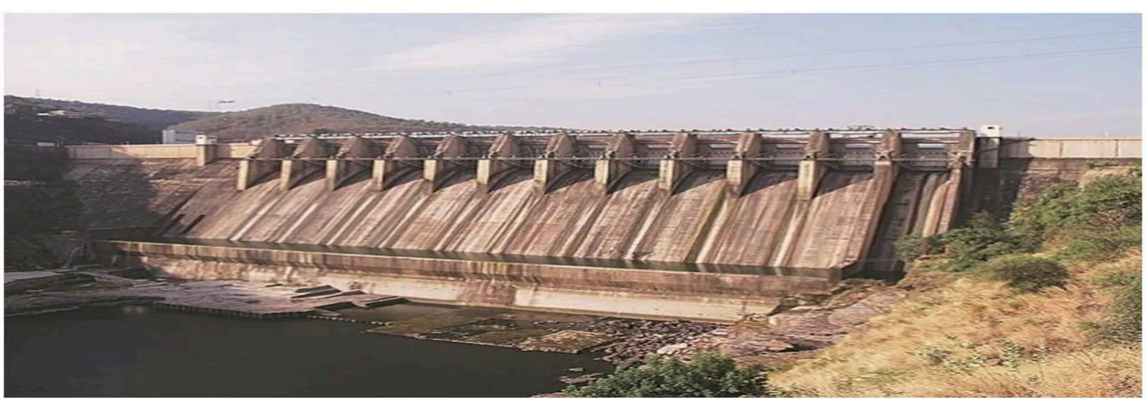
- These include planting vegetation to retain extra water.
- Terracing hillsides to slow flow downhill.
- Construction of floodways (man-made channels to divert floodwater).
- Other techniques include the construction of levees, lakes, dams, reservoirs, retention ponds to hold extra water during times of flooding.

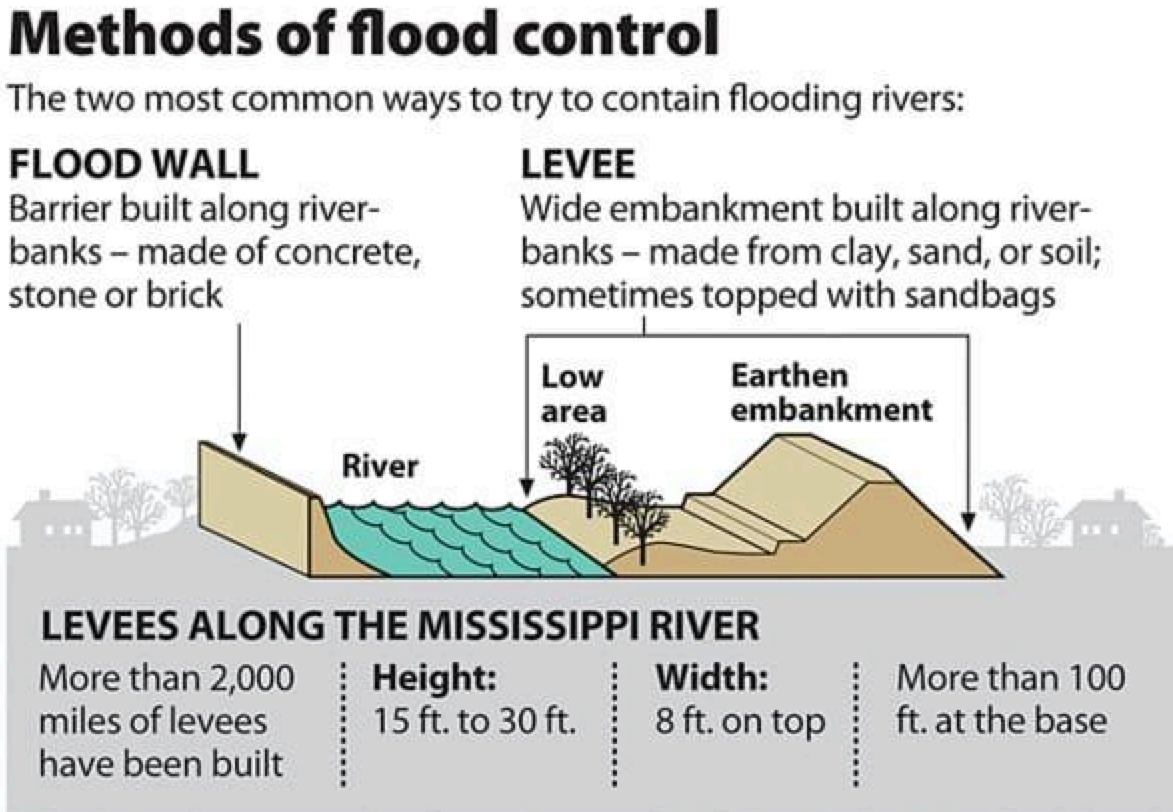
COSTAL MANEGEMENT

Coastal management is barrier against flooding and erosion, and techniques that stop erosion to claim lands. Protection against rising sea levels in the 21st century is crucial, as sea level rise accelerates due to climate change. Changes in sea level damage beaches and coastal sediments to be disturbed by tidal energy.

DAMS

- Many dams and their associated reservoirs are designed completely or partially to aid in flood protection and control.
- Many large dams have flood control reservoirs in which the level of reservoirs must be kept below a certain elevation before the onset of the rainy/summer melt season to allow a certain amount of space in which floodwaters can fill.





Flood control methods are used to prevent the detrimental effects of flood waters.

- Flood relief methods used to reduce the effects of flood water or high water level. Flooding can be caused by a mix of both natural processes, such as extreme weather upstream, and human changes to water bodies and runoff.
- Building hard infrastructure to prevent flooding, such as flood walls, can be effective at managing flooding increased best practice within landscape engineering is to rely more on soft infrastructure and natural system.

- For flooding on coasts, coastal management practices have to not only handle changes water flow, but also natural processes like tides.

How can we prevent flood for future?

1. **Introduce better flood warning system**.

Improve our flood warning systems, giving people more time to take action during flooding, potentially saving lives.

2. **Modify homes and businesses to help them withstand floods**

The focus should be on ‘flood resilience’. Concreting floors and replacing materials such as MDF and plasterboard with more robust alternatives. Waterproofing homes and businesses and moving electric sockets higher u the walls to increase resilience.

3. **Construct building above flood levels**

Construct new buildings one meter from the ground to prevent flood damage. Conventional activities had to be supplemented with more innovative methods to lower the risk of future disasters.

4. **Tackle climate changes**

Climate change has contributed to rise in extreme weather events, scientists believe. ’Pursue efforts’ to limit the increase in the global average temperature above preindustrial levels.

5. **Protect wetland and introduce plant trees**

The creation of more wetland – which can at as sponges, soaking up moisture – and wooded areas can slow down waters when rivers overflow. These areas are often destroyed to make room for agriculture and development. Halting deforestation and wetland drainage, reforesting upstream areas and restoring damaged wetlands could significantly reduce the impact of climate change on flooding, according to the conservation charity.

6. Restore rivers to their natural courses

Many river channels have been historically straightened to improve navigability. Remaindering straightened rivers by introducing their bends once more increases their length can delay the flood flow and reduce the impact of the flooding downstream

7. Improve soil conditions

Inappropriate soil management, machinery and animal hooves can cause soil to become compacted so that instead if absorbing moisture, holding it and slowly letting it go, water runoff it immediately. Well drained soil can absorb huge quantities of rainwater, preventing it from running into rivers.

8. Put up more flood barriers

Temporary barriers are created to permanent flood protection, such as raised embankments, increasing the level of protection.